

Negative energies states in TTBM

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It is shown that within TTBM — a recent deterministic theory in which the apparently paradoxical aspects of Quantum Mechanics are the effect of the existence of an extra unobservable time — negative energy states represent particles that do not respect the uncertainty principle and have therefore never been observed. This property could be interpreted as detrimental to the theory, but it is nevertheless compatible with the symmetric theory proposed by Majorana in 1937 and still having to be tested by experience.

Keywords: Quantum Mechanics Foundations; Negative energies; Antimatter; Majorana's symmetric theory.

I. INTRODUCTION

TTBM [1–3] arises as a result of literally taking Feynman's sum over history interpretation [4–8]. Determinism is restored through the introduction of extra hidden dimension of time, called τ . Movements in τ , caused by a Bohmian-type quantum potential, are instantaneous with respect to the usual time and, while not directly observable, are supposed to generate the quantum uncertainties of the observables. For a particle with *positive* energy, the famous *Zitterbewegung* motion [9, 10] is deduced and a more general law of it, free from the call of antimatter, is achieved. However, the case of a particle with *negative* energy has not yet been studied: what motion does the quantum potential generate in that case?

The following section presents the fundamental concepts and basic equations of the TTBM theory and the next one studies the case of negative-energy free states. A final section discusses the results, comparing them with the standard theory and with the one proposed by Majorana in 1937, which still awaits experimental verification.

II. BASIC TTBM EQUATIONS

In TTBM there are two velocities, one *classic* $\vec{v}_c$ and one *intrinsic* v_i , which is *omnidirectional*, that is, with a non-zero component in every direction $\hat{s}$. If $S(t, \tau)$ is the generic coordinate of the particle, we have

$$S(t, \tau) = S_{t_0} + S_{\tau_0} + \int_{t_0}^t v_{c_s}(\xi) d\xi + \int_{\tau_0}^{\tau} v_{i_s}(\zeta) d\zeta. \quad (1)$$

This equation allows to the particle, observed in its trajectory as a function of t , to jump instantaneously, through the time dimension τ , in any direction towards an arbitrarily distant point in space. This is in accordance with a *literal* interpretation of the Feynman's sum over history formulation, in which the physical event "the particle (or photon) moves from A to B " is only correctly explained by admitting that it has traveled *all the possible paths from A to B at once*, also interacting with every

possible virtual particle of the vacuum [4–8]. The basic equations of TTBM are obtained from the wavefunction

$$\psi(\vec{r}, t, \tau) = R(\vec{r}, t, \tau) e^{\frac{i}{\hbar} S(\vec{r}, t, \tau)}, \quad (2)$$

with $R > 0$ and S real functions, admitting the validity of the Klein-Gordon equation (generalized if a scalar potential V is present). For the general case $V \neq 0$ one gets [3]:

$$(H - V)^2 = m^2 \gamma_c^2 c^4 + \hbar^2 \left(\frac{1}{R} \frac{\partial^2 R}{\partial t^2} - c^2 \frac{\nabla^2 R}{R} \right), \quad (3)$$

$$\frac{\partial(R^2(H - V))}{\partial t} + c^2 \vec{\nabla} \cdot (R^2 \vec{p}_c) = 0, \quad (4)$$

having assumed $\vec{\nabla} S = \vec{p}_c = m \gamma_c \vec{v}_c$ and $\frac{\partial S}{\partial t} = -H$. If we set $H \equiv m \gamma_c c^2 + V + V_Q$ to define the Quantum Potential V_Q , eq. (3) attests that it is not zero in general.

Eq. (4) is nothing other than the well-known continuity equation that can also be obtained by applying the Klein-Gordon equation to ψ and ψ^* and which fails to represent the conservation of a probability density (see, e.g. [11]). Despite this, in TTBM *the non-relativistic meaning of R^2 as volumetric probability density is maintained*. For that, it is enough to *expressly* request the validity of the continuity equation:

$$\frac{\partial R^2}{\partial t} + \vec{\nabla} \cdot (R^2 \vec{v}_c) = 0, \quad (5)$$

which imposes, also in Relativity, the conservation of the probability of finding the particle in a unit volume. The novelty is that, if R (and therefore also V_Q) depends on t time, as TTBM assumes, eqs. (5) and (4) are not incompatible, but together give

$$R^2 \frac{d(m \gamma_c c^2)}{dt} + \frac{\partial(R^2 V_Q)}{\partial t} = 0, \quad (6)$$

which relates the temporal variation of the quantum potential to that of the relativistic mass.

The last equation required by the model is a wave equation for the generic τ -motion:

$$\frac{\partial R^2}{\partial \tau} - v_{i_s} \frac{\partial R^2}{\partial s} = 0, \quad (7)$$

for any direction $\hat{s}$. Its justification is consequent to the uncertainty generator function that the model assigns to the intrinsic motions [2].

The motion of the particle is decomposed into a classical and a purely quantum part. The law of classical motion is well known: if V is conservative, the classic total energy $m\gamma_c c^2 + V$ is constant and one gets

$$\frac{d\vec{p}_c}{dt} = -\vec{\nabla}V. \quad (8)$$

As for intrinsic motions, it can be shown that it does not obey Special Relativity [2]. Then, since the mass experiences only the relativistic increase due to $\vec{v}_c$, from the conservation of energy for the generic τ -motion, one deduces

$$m\gamma_c \frac{dv_{i_s}}{d\tau} = -\frac{\partial V_Q}{\partial s}. \quad (9)$$

III. FREE PARTICLE WITH NEGATIVE ENERGY

For a free particle, eq. (8) informs us that $\vec{p}_c$ is constant. Retracing the same steps taken in [2], we particularize eq. (3) for an arbitrary direction $\hat{s}$, using a polar coordinate system (s, θ, ϕ) , where θ is the angle between $\hat{s}$ and $\hat{v}_c$ and ϕ the azimuth. It can be obtained

$$H = \pm \sqrt{m^2 \gamma_c^2 c^4 - \hbar^2 \frac{c^2}{\gamma_{c_s}^2} \frac{R''}{R} - \frac{2\hbar^2 c^2}{sR} R'} \quad (10)$$

where γ_{c_s} refers to $v_{c_s} \equiv v_c \cos \theta$ and the apostrophe indicates derivative over s . On the other hand, making use of continuity eq. (5), eq. (4) becomes

$$v_{c_s} \frac{R'}{R} (m\gamma_c c^2 - H) + \frac{1}{2} \frac{\partial H}{\partial t} = 0, \quad (11)$$

in which we continue to assume for R a bell shape with maximum R_M . These equations have been solved in the neighborhood of R_M for $H > 0$, finding a quadratic V_Q which, through eq. (9), gives rise to a harmonic τ -oscillation, which causes quantum indeterminacy.

Now we are interested in the minus sign in eq. (10). First of all, as already done in [2], we neglect the last addend of eq. (10), which would be small in the neighborhood of the maximum R_M . Then, placing $\lambda^2 \equiv \frac{\hbar^2}{m^2 c^2 \gamma_{c_s}^2}$, one can get, by substitution of eq. (10) into eq. (11):

$$v_{c_s} \frac{R'}{R} (\sqrt{\beta} + \beta) = \frac{1}{4} \frac{\partial \beta}{\partial t}, \quad (12)$$

where $\beta \equiv 1 - \lambda^2 \frac{R''}{R}$. Since, for constant $\vec{v}_c$, the continuity eq. (5) holds for any function of any-order derivative of R over t or space, we can apply it to β , obtaining $\frac{\partial \beta}{\partial t} = -\beta' v_{c_s}$. Substituting:

$$\frac{R'}{R} = -\frac{1}{4} \frac{\beta'}{\sqrt{\beta} + \beta}. \quad (13)$$

Multiplying by ds and integrating member by member one obtains

$$-1 + \frac{c_1^2}{R^2} = \sqrt{\beta}, \quad (14)$$

with c_1 arbitrary constant. Since $\sqrt{\beta} = \frac{-H}{m\gamma_c c^2}$, we finally have

$$H = m\gamma_c c^2 \left(1 - \frac{c_1^2}{R^2}\right). \quad (15)$$

This equation shows that the negative energy state is in any case associated with a *positive* mass, with a quantum potential given by $V_Q = -m\gamma_c c^2 \frac{c_1^2}{R^2}$. In fact, even if one wants to assume $H = -m\gamma_c c^2 + V_Q$, from eq. (15) it would result $V_Q = 2m\gamma_c c^2 - m\gamma_c c^2 \frac{c_1^2}{R^2}$, which returns a positive value for the *effective* mass and differs by an insignificant constant from $-m\gamma_c c^2 \frac{c_1^2}{R^2}$. On the other hand, eq. (6) confirms that $V_Q = \frac{k}{R^2}$, with k constant. Placing $k \equiv m\gamma_c c^2 R_M^2 f$, where f is an adimensional negative constant, we get

$$V_Q = m\gamma_c c^2 f \frac{R_M^2}{R^2}, \quad (16)$$

where the condition $H \leq 0$ implies $f \leq -1$.

Now let's solve

$$(1 + f \frac{R_M^2}{R^2})^2 = 1 - \lambda^2 \frac{R''}{R}, \quad (17)$$

in which the variable s is missing. One gets

$$R' = \pm \frac{R_M \sqrt{-f}}{\lambda} \sqrt{4 \ln \frac{R}{R_M} + f(1 - \frac{R_M^2}{R^2})} \quad (18)$$

where the sign $+$ should be for $s < s_M$ and vice versa. Separating the variables:

$$\int \frac{d(\frac{R}{R_M})}{\sqrt{4 \ln \frac{R}{R_M} + f(1 - \frac{R_M^2}{R^2})}} = -\frac{\sqrt{-f}}{\lambda} |s - s_M|, \quad (19)$$

which cannot be solved exactly. For $R \rightarrow R_M$, one can however obtain

$$R \simeq R_M \left(1 - \frac{f(f+2)}{2\lambda^2} (s - s_M)^2\right), \quad (20)$$

which only for $f < -2$ can represent a bell-shaped curve on the maximum R_M . Since

$$\begin{aligned} \frac{1}{R^2} &\simeq \frac{1}{R_M^2} \frac{1}{1 - \frac{f(f+2)}{\lambda^2}(s - s_M)^2} \simeq \\ &\simeq \frac{1}{R_M^2} \left(1 + \frac{f(f+2)}{\lambda^2}(s - s_M)^2\right), \end{aligned} \quad (21)$$

we have finally

$$V_Q \simeq m\gamma_c c^2 f \left(1 + \frac{f(f+2)}{\lambda^2}(s - s_M)^2\right). \quad (22)$$

Eq. (9) then gives rise to the following solution for the abscissa S of the particle:

$$S(\tau) - S_M \simeq c_1 e^{\sqrt{k}\tau} + c_2 e^{-\sqrt{k}\tau}, \quad (23)$$

where $k \equiv -\frac{2c^2 f^2(f+2)}{\lambda^2}$ is certainly positive for $f < -2$. To avoid divergence all that remains is to set $c_1 = 0$, obtaining a solution that rapidly tends to zero. For the mass of an electron, for example, assuming $\sqrt{k} \approx \frac{c}{\lambda_c}$, we obtain an extinction time of the order of 10^{-21} s; for a particle of greater mass this time is even shorter.

In conclusion, in TTBM, a sufficiently negative energy state can be associated with a positive-mass particle with rapidly vanishing τ -motions; this means that, according to the hypotheses the TTBM theory, it would not possess quantum indeterminacy. For such particles, the dynamics would be rigorously classical, without respect for the uncertainty principle.

IV. DISCUSSION

Particles of this type, if they exist, have not yet been observed: all known particles (including antimatter [12–14]) obey the principles of quantum mechanics. But the

real flaw for TTBM would be that it is incapable to describe the negative-energy states that standard Quantum Field Theory (QFT) associates with antiparticle creation operators [15, 16]. However, this interpretation has not yet been proven to be necessary. A symmetric theory proposed by Majorana in 1937 reinterprets these states as positive-energy states with opposite charges [17]. Only by a curious fate, linked to an unfortunate neglect of Majorana's work, this theory is still considered an exotic hypothesis today, when in fact it is a simpler model with several advantages that are certainly worth reiterating [18–20]:

1. By losing the need for antimatter to be something intrinsically different from matter, observed reality is halved.
2. The Dirac equation can be derived by exploiting the variational principle starting from a suitable Lagrangian.
3. The previous deduction also yields the correct fermionic statistics; while in QFT it must be added "by hand".
4. The above advantages do not detract the modus operandi of QFT, which continues to function normally.

Even considering only the first point, it seems that Occam's razor has not yet been applied to the standard viewpoint; although, certainly, the experiments must have the last word. If Majorana's theory is correct, then antimatter is an ultimately useless concept, since an antiparticle can be considered a normal particle with opposite charges; in particular, neutrinos and antineutrinos would coincide. TTBM would be perfectly integrated into this model. As for its prediction of particles that have not yet been observed, this is certainly not an inconsistency, since nothing forces everything mathematically possible to exist. And, however, experiments could reveal surprises.

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